

Reconstruction of Black Body Spectrum through Numerical Methods and Spline Interpolation

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A computational study of black body radiation spectrum based on Planck's Law is presented. The main objective is to generate synthetic spectra for different characteristic temperatures and quantitatively analyze the position of the maximum emission wavelength using numerical methods. Python routines were implemented to evaluate Planck's Law, perform numerical integration, reconstruct spectra, and locate the maximum through dense sampling and cubic spline fitting. This work constitutes an advance towards the computational understanding of black body physics and provides foundations for developing spectral analysis practices.

INTRODUCTION

The electromagnetic radiation emitted by a black body was one of the historical problems that drove the development of quantum mechanics. At the end of the 19th century, discrepancies between the Rayleigh–Jeans and Wien laws motivated Planck to propose his quantum description of the spectral distribution of energy [1], establishing the foundations of modern quantum theory [2].

Currently, Planck's law remains fundamental in astrophysics, spectroscopy, and characterization of thermal sources. Numerical simulation of spectra allows analyzing temperature dependence and validating derived laws such as Wien's displacement law and the Stefan–Boltzmann law. Various studies address spectral reconstruction of thermal radiation through experimental and computational techniques, reinforcing its relevance in scientific and technological applications. In particular, recent works have explored spectral reconstruction methods using advanced computational approaches [3, 4], while others have employed Planck's law as the main reference in radiometric calibration processes [5]. These precedents allow framing the present study as a formative and computational approximation to these classical physics problems.

In this work, synthetic black body spectra are generated for different temperatures through numerical methods and cubic spline interpolation techniques, in order to visualize and understand the spectrum evolution in a quantitative and accessible manner. Computational routines are developed in Python that allow evaluating Planck's law, reconstructing the continuous spectrum from discrete data, and precisely locating the maximum emission wavelength, providing a didactic and methodological tool for spectral analysis in computational physics.

THEORETICAL BACKGROUND

Planck's Law.

The spectral irradiance per unit wavelength of a black body at temperature T is given by

$$B_\lambda(\lambda, T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}, \quad (1)$$

where h is Planck's constant, c the speed of light, and k_B Boltzmann's constant. This expression resolves the ultraviolet catastrophe of Rayleigh–Jeans and correctly describes the spectrum over the entire wavelength range.

Wien's Displacement Law.

The maximum of the spectral distribution satisfies

$$\lambda_{\max} T = 2.898 \times 10^{-3} \text{ m}\cdot\text{K}. \quad (2)$$

This relationship is obtained by maximizing Planck's function and allows computational validation of the generated spectra. At higher temperatures, the peak shifts towards shorter wavelengths.

Stefan–Boltzmann Law.

The total power per unit area results from integrating the complete spectrum:

$$j^\star = \sigma T^4, \quad (3)$$

where $\sigma = 5.670 \times 10^{-8} \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-4}$. The quartic dependence implies that small increments in T dramatically increase the radiated power.

COMPUTATIONAL DETAILS

Definition of spectral domain.

The range $\lambda \in [200 \text{ nm}, 3000 \text{ nm}]$ is defined, from the near ultraviolet to the mid-infrared, uniformly discretized with $N \approx 1000\text{--}2000$ points to capture the complete spectral shape.

Numerical evaluation.

Planck's law is evaluated point by point using Python and NumPy, generating vectors that represent $B_\lambda(\lambda, T)$ at each wavelength of the discretized domain.

Cubic spline interpolation.

Interpolation was implemented using cubic splines (`scipy.interpolate`) that guarantee continuity up to the second derivative. This allows evaluating the spectrum at arbitrary wavelengths, precisely locating the maximum through dense sampling, and numerically integrating with high accuracy.

Maximum localization.

$\lambda_{\max}$ is determined through dense sampling ($\sim 10,000$ points) of the spline in the region predicted by Wien, reaching precisions on the order of nanometers. The value is compared with the theoretical prediction to quantify the error.

Numerical validation.

Total power is calculated by integrating the spectrum with the trapezoidal rule (`numpy.trapz`) and comparing with Stefan-Boltzmann. Spectra from 3000 K to 6000 K were analyzed, verifying maximum displacements and the T^4 dependence of power. Visualization with `matplotlib` allows analyzing spectral variation, peak displacement, and irradiance increase.

RESULTS

The results obtained from the computational experiments described in the methodology are presented here. For each experiment, the spectrum reconstruction by different methods, the estimation of the spectral maximum position ($\lambda_{\max}$), the temperature estimated from Wien's law, the partial integral of the spectrum (power in the measured range), and the error percentages calculated according to theoretical values are shown.

Three representative experiments were performed:

Experiment A: Main case.

The reference case is considered with temperature $T = 5800$ K, corresponding approximately to the Sun's surface temperature. The spectral domain is defined in the range $\lambda \in [0.1, 3.0] \mu\text{m}$ with $N = 20$ uniformly distributed sampled points. Random Gaussian noise with relative amplitude $\alpha = 5\%$ is added to the generated synthetic data to simulate realistic experimental conditions.

Experiment B: Sampling sensitivity.

To analyze the effect of point density on reconstruction, the temperature $T = 5800$ K and the same spectral range as the main case are maintained, but the number of sampled points is systematically varied considering $N \in \{10, 30, 100\}$. In all cases, relative noise $\alpha = 5\%$ is incorporated to maintain comparable conditions with experiment A.

Experiment C: Noise sensitivity.

This experiment evaluates the robustness of reconstruction methods under different levels of experimental uncertainty. A lower temperature $T = 3000$ K is used with $N = 30$ sampled points, varying the noise amplitude at three levels: $\alpha \in \{1\%, 5\%, 10\%\}$. This configuration allows quantifying how reconstruction quality degrades as noise increases.

For each of the described experiments, three different reconstruction methods were implemented and compared. The first method corresponds to the interpolating cubic spline, which passes exactly through all data points and guarantees continuity up to the second derivative. The second method uses smoothing splines, which do not necessarily pass through all points but minimize a combination between data fitting and curve smoothness; the smoothing parameter s is selected heuristically for each case. The third method consists of a direct parametric fit to the functional form of Planck's Law, called "Planck-fit", in which a global scale factor is optimized to minimize the quadratic error with respect to the data.

Results: Experiment A (Reference case)

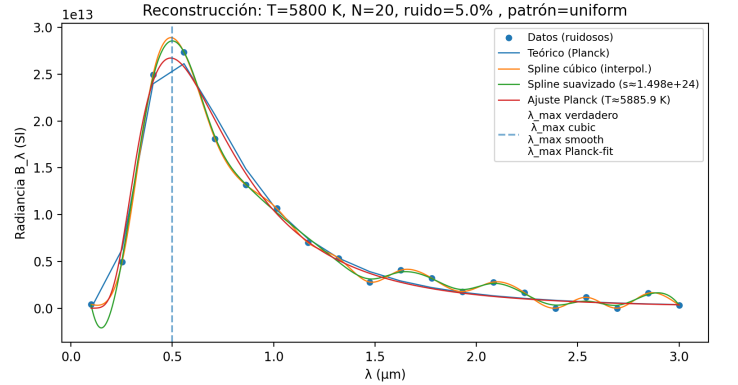


FIG. 1. Spectrum reconstruction for Experiment A ($T = 5800$ K, $N = 20$, 5% noise). Points: noisy sampled data; black line: theoretical spectrum; colored lines: reconstructions by cubic spline, smoothing spline, and Planck fit.

Maximum position:

Compare the estimated $\lambda_{\max}$ with the theoretical value $\lambda_{\max}^{\text{true}} = b/T$; the percentage divergence indicates bias due to sampling and noise. In general, we expect that with $N = 20$ and 5% noise, the smoothing spline reduces variance compared to the interpolating spline, while the Planck fit provides a relatively stable estimate of T if the calibration (factor A) compensates for scale differences.

Partial area:

The integral of the interpolating spline reproduces the central part of the spectrum; the difference with the partial theoretical integral reflects information loss in the

tails and the effect of noise. **RMSE:**

It is a measure of fidelity at sampled points; lower values indicate better local fit.

Results: Experiment B (Variation of number of points N)

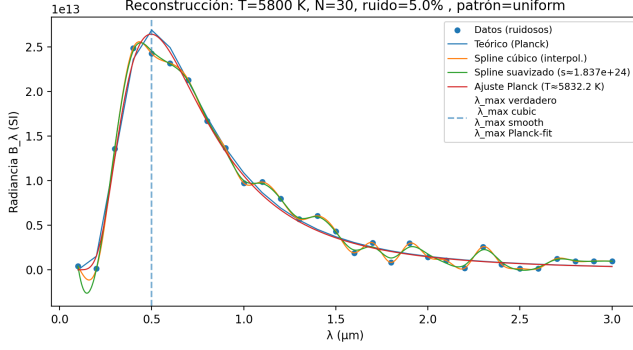


FIG. 2. Comparison of reconstructions for different N (Experiment B). Observe the improvement in peak localization and shape as N increases.

As the number of sampled points N increases, a systematic reduction is systematically observed in both the error in spectral peak localization and the root mean square error (RMSE) in the global spectrum reconstruction.

This behavior is expected since higher point density provides more information about the curve shape. However, for small values of N (for example, $N = 10$), the interpolating cubic spline may overfit the random noise fluctuations, producing unstable estimates of $\lambda_{\max}$ and spurious oscillations between data points.

In these cases, the smoothing spline offers a more robust alternative by reducing estimation variance, although this occurs at the cost of introducing a slight systematic bias in the maximum position. For its part, the direct parametric fit to Planck's form may present convergence difficulties or generate considerable uncertainties when the sampled points do not adequately cover both the peak region and the spectrum tails, limiting its applicability in sparse sampling regimes.

Results: Experiment C (Noise variation)

As expected, the root mean square error (RMSE) of the reconstruction grows monotonically with the noise level in the data.

Among the evaluated methods, the interpolating cubic spline shows the highest sensitivity to noise, manifesting in high variance in both the reconstructed spectrum shape and the estimation of the maximum position. The

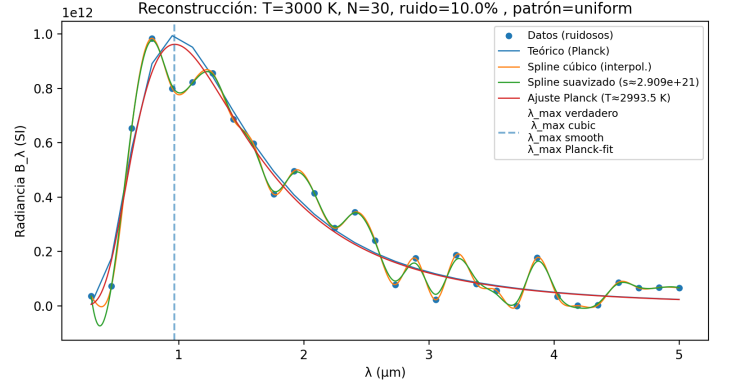


FIG. 3. Effect of noise level on method precision (Experiment C).

smoothing spline significantly mitigates this problem, reducing both the global RMSE and the statistical dispersion in the determination of $\lambda_{\max}$.

However, this gain in stability has a cost: if the smoothing parameter s is chosen too large, a systematic bias is introduced that slightly displaces the estimated peak position from its true value.

For its part, the direct parametric fit to Planck's Law demonstrates remarkable robustness in temperature T estimation when the theoretical model is correct and the signal-to-noise ratio (SNR) remains at reasonable levels. However, under high noise conditions, the numerical optimization may present convergence problems, causing the method to fail or provide physically inconsistent results.

Global comparison of methods

Table I allows comparing the performance of spectral reconstruction methods in terms of spectral maximum estimation, inferred temperature, and total emitted power. Under adequate sampling conditions and low noise, all methods reasonably reproduce theoretical values, although the direct fit to Planck's law presents the highest global agreement. Interpolation methods using splines show small deviations in maximum localization, associated with the sensitivity of numerical differentiation.

When reducing the number of samples or increasing the noise level, interpolation methods are more affected, with the cubic spline being the most sensitive and the smoothing spline the most stable. In contrast, the Planck fit maintains consistent estimates in all cases. For total emitted power, all methods correctly reproduce the expected trend, showing that numerical integration is less sensitive to local spectrum fluctuations.

Experiment		A	B				C		
N (Samples)		20	10	30	100		30	30	30
Noise (α)		0.050	0.050	0.050	0.050		0.010	0.050	0.100
λ_{max} [μm]	Theoretical (λ_{theor})	0.500	0.500	0.500	0.500		0.966	0.966	0.966
	Cubic (λ_c)	0.493	0.514	0.433	0.489		0.948	0.824	0.786
	Smooth (λ_s)	0.500	0.510	0.440	0.485		0.987	0.872	0.788
	Planck (λ_p)	0.492	0.510	0.497	0.502		0.966	0.967	0.968
T_{wien} [K]	Theoretical (T_{theor})	5800	5800	5800	5800		3000	3000	3000
	Cubic (T_c)	5,874.233	5,635.305	6,697.910	5,928.834		3,058.190	3,517.574	3,688.666
	Smooth (T_s)	5,793.922	5,683.119	6,579.493	5,973.115		2,937.034	3,322.659	3,677.645
	Planck-fit (T_p)	5,885.899	5,684.535	5,832.211	5,767.705		2,999.416	2,996.978	2,993.510
Area partial	Theoretical (A_{theor})	2.00×10^7	2.00×10^7	2.00×10^7	2.00×10^7		1.42×10^6	1.42×10^6	1.42×10^6
	Cubic (A_c)	2.00×10^7	2.03×10^7	2.00×10^7	2.01×10^7		1.42×10^6	1.42×10^6	1.42×10^6
	Smooth (A_s)	1.99×10^7	2.06×10^7	1.99×10^7	2.01×10^7		1.41×10^6	1.41×10^6	1.41×10^6
	Planck-fit (A_p)	1.96×10^7	1.99×10^7	1.96×10^7	2.02×10^7		1.41×10^6	1.39×10^6	1.37×10^6

TABLE I. Results of spectral reconstruction for different experiments, showing the estimation of spectral maximum, inferred temperature, and partial emitted power, and allowing comparison of the effect of sampling, noise, and numerical method employed.

Calculated Error Analysis:

%Error	Experiment	A	B				C		
λ_{max}	Cubic (λ_c)	1.26	2.92	13.41	2.17	1.90	14.71	18.67	
	Smooth (λ_s)	0.10	2.06	11.85	2.90	2.14	9.71	22.66	
	Planck-fit (λ_p)	1.46	2.03	0.55	0.56	0.02	0.10	0.27	
T_{wien}	Cubic (T_c)	1.28	2.84	15.48	2.22	1.94	17.25	22.96	
	Smooth (T_s)	0.10	2.02	13.44	2.98	2.10	10.76	22.59	
	Planck-fit (T_p)	1.48	1.99	0.56	0.56	0.02	0.10	0.22	
Area	Cubic (A_c)	0.09	1.35	0.08	0.54	0.07	0.05	0.06	
	Smooth (A_s)	0.72	3.10	0.47	0.50	0.15	0.12	0.33	
	Planck-fit (A_p)	1.91	0.35	2.21	1.09	0.31	1.55	3.10	

TABLE II. Relative error percentage for λ_{max} , T_{wien} and partial area.

Table II shows that the largest errors are concentrated in the estimation of λ_{max} and T_{wien} , while the calculation of partial area is more stable. Interpolation methods show strong dependence on sampling and noise, with the cubic spline being the most sensitive and the smoothing spline the most stable, tending to produce more concentrated distributions than the interpolating spline. In contrast, the Planck's law fit exhibits the smallest errors in the estimation of λ_{max} and temperature, remaining robust against variations in data quality.

Physical discussion

The results show that spline-based reconstruction allows recovering the general shape of a black body spectrum even with a moderate number of points ($N \gtrsim 20$). The position of the maximum, used to estimate

T through Wien's law, is sensitive to sampling density and noise; improving N and densifying around the peak reduces the error. In summary, for a study it is recommended to present the combination: smoothing spline to obtain a robust reconstruction and Planck fit as parametric verification of the estimated temperature.

CONCLUSIONS

In this work, a simple and consistent computational study was developed for the generation and reconstruction of black body radiation spectra, based on Planck's law and numerical methods accessible at the undergraduate level. The obtained results correctly reproduce the fundamental laws associated with the phenomenon, such as Wien's displacement law and the Stefan-Boltzmann law, validating the physical coherence of the implemented approach.

The comparison between methods showed that spline interpolation techniques are suitable for well-sampled data with low noise level, while the smoothing spline improves stability against experimental fluctuations. The direct fit to Planck's law proved to be the most robust and precise method for temperature estimation, even under limited sampling conditions or higher noise.

Overall, the study shows that the computational methods employed allow efficient analysis of local and global properties of the black body spectrum, constituting a useful formative tool for the study of thermal and spectral phenomena.

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